Roll No.*:_____

Thapar Institute of Engineering and Technology, Patiala
Department of Computer Science & Engineering
Mid Semester Test (MST - 15/03/2024)

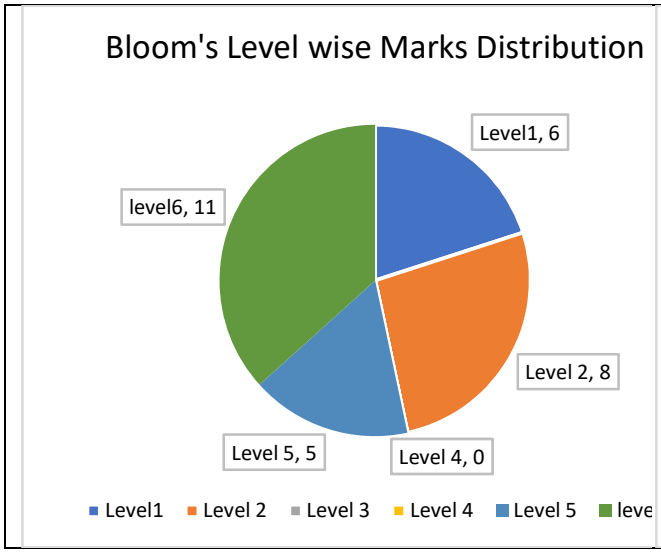
MSc. Ist year(Semester-II) **PMC212: Data Structures and Algorithms**
Time: 2 hrs; MM:30 **Faculty Coordinator: Palika Chopra**

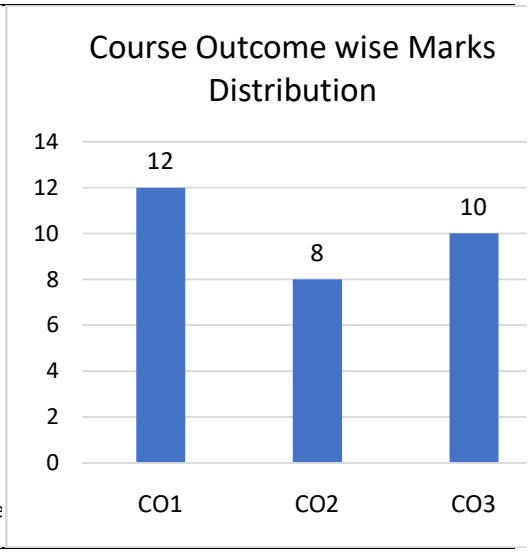
Note: All questions are compulsory. Pencil work will not be evaluated.

Q.No	Questions	Marks	CO	BL
Q1:	Consider the arithmetic expression P, written in postfix notation: **P: 12, 7, 3, -, /, 2, 1, 5, +, *, +** (a) Evaluate the expression 'P' using a stack. Visualize all the intermediate steps in the stack. (b) Translate the expression 'P' into its equivalent infix expression.	(6)	CO1	L3
Q2:	Write a C++ program to sort the provided list of numbers using the Bubble Sort algorithm. Provide a step-by-step visualization of each iteration of the sorting process. **Input:** List of numbers to sort: {5, 3, 8, 2, 1} **Output:** Sorted list: {1, 2, 3, 5, 8} **Complexity:** Discuss the time complexity of the **Bubble Sort algorithm** in terms of its worst-case, best-case, and average-case scenarios. Additionally, comment on its space complexity and any potential drawbacks associated with its performance. **Comparison:** Furthermore, consider the possibility of achieving the same sorting outcome using the **Quick Sort algorithm**. Discuss how Quick Sort's time and space complexity compare to Bubble Sort and highlight any advantages or disadvantages of using Quick Sort in this context.	(8)	CO2	L2
Q3:	Write an algorithm named LINK_LIST, to create a linked list where each node can store a group of four digits from a given number. For instance, if the number is **23156782913287**, the linked list should represent the number as follows: **3287 -> 8291 -> 1567 -> 23.**	(5)	CO3	L6
Q4:	a) Explain the difference between an array and a linked list in terms of memory allocation and access time. b) Differentiate between linear search and binary search in terms of their time complexity and applicability to sorted and unsorted lists.	(6)	CO1	L1

Q5:	Write a C++ program to compute the nth term of the harmonic series using recursion. The harmonic series is defined as:
	
	$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}$$
	
	The program should prompt the user to enter the value of n. After receiving the input, it should compute and print the nth term of the harmonic series.
	Here's a sample run of the program:
	
	`Enter the value of n: 5`
	`The 5th term of the harmonic series is: 2.283333`
		(5)	CO3	L5

*****All The Best*****


Bloom's Level wise Marks Distribution
Level1, 6
level6, 11
Level 2, 8
Level 5, 5
Level 4, 0
Level1
Level 2
Level 3
Level 4
Level 5
leve


Course Outcome wise Marks Distribution
14
12
10
8
6
4
2
0
12
8
10
CO1
CO2
CO3